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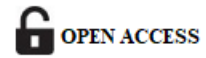
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BODY FAT PERCENTAGE AS AN INDEPENDENT PREDICTOR OF FUNCTIONAL MOVEMENT QUALITY

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ABSTRACT

Background: Excess adiposity is associated with cardiometabolic disease, musculoskeletal dysfunction, and altered biomechanical efficiency¹⁻⁵. However, composition-specific relationships between adiposity and functional movement competency remain incompletely characterized in adult populations.

Purpose: To determine whether DXA-derived body fat percentage independently predicts functional movement quality and to evaluate differential sensitivity across discrete movement tasks.

Methods: Cross-sectional analysis of 265 adults aged 18–65 years. Body composition was assessed via dual-energy X-ray absorptiometry. Functional movement was evaluated using deep squat and posterior rocking assessments scored on a 1–3 ordinal scale. Pearson correlations and multivariable regression adjusting for age and sex were conducted.

Results: Body fat percentage was inversely associated with composite movement score ($r = -0.417$, $p < 0.001$). Deep squat demonstrated stronger association ($r = -0.391$) than posterior rocking ($r = -0.069$). In adjusted regression, body fat percentage remained independently predictive ($\beta = -0.0117$, 95% CI -0.0155 to -0.0080), explaining 0.208 of variance.

Conclusion: *Adiposity independently predicts functional movement quality, with multi-joint closed-chain tasks demonstrating greater biomechanical sensitivity.*

Keywords: Body Fat Percentage; Functional Movement; DXA; Adiposity; Movement Quality.

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Introduction

Obesity and excess adiposity represent critical determinants of altered human movement and musculoskeletal strain¹⁻⁵. Although the cardiometabolic implications of elevated body fat are extensively documented, the biomechanical and neuromotor consequences remain comparatively underexplored in community populations. Increased adiposity modifies joint loading patterns, increases compressive and shear forces, shifts center-of-mass distribution, and reduces relative strength capacity¹⁰⁻¹⁴. These alterations may impair the integrated coordination required for efficient multi-joint movement execution.

Functional movement competency reflects the interaction of mobility, stability, proprioception, and neuromuscular control⁶⁻⁹. Closed-chain tasks such as the deep squat demand synchronized torque generation across the hip, knee, ankle, and trunk while maintaining dynamic postural equilibrium. Posterior rocking emphasizes lumbopelvic control under comparatively lower axial load. Therefore, differential task sensitivity to adiposity may reveal mechanistic gradients in biomechanical constraint.

Body mass index (BMI), though widely used, does not distinguish between lean and fat mass and may obscure composition-specific associations¹⁵⁻¹⁷. Dual-energy X-ray absorptiometry (DXA) provides gold-standard quantification of total body fat percentage, enabling precise examination of adiposity-driven biomechanical influences. The present study investigates whether DXA-derived adiposity independently predicts movement competency and evaluates which discrete movement patterns demonstrate the strongest association.

Methods

Participants completed standardized DXA body composition assessment and functional movement testing. Deep squat and posterior rocking assessments were scored 1–3 by trained evaluators. A composite score represented integrated movement competency. Descriptive statistics characterized the sample.

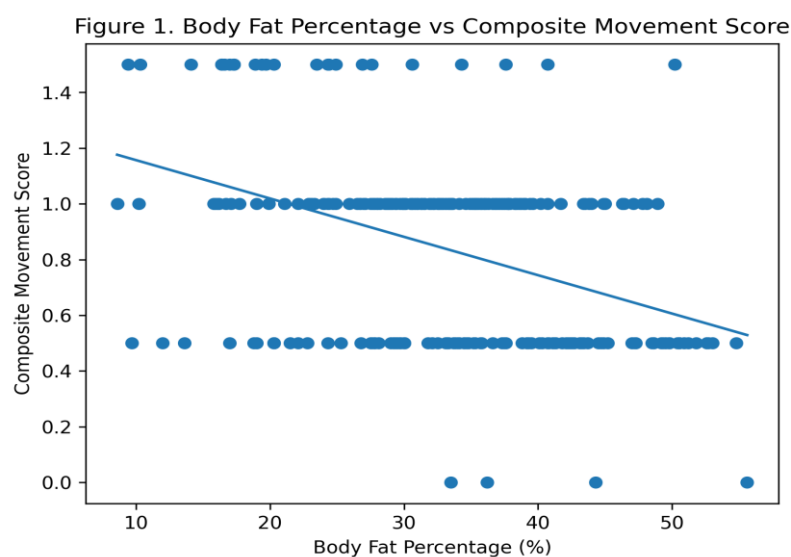
Pearson correlation coefficients quantified bivariate relationships between body fat percentage and each movement outcome. Multivariable ordinary least squares regression adjusted for age and sex determined independent predictive capacity. Effect magnitude comparisons were performed between tasks to evaluate relative adiposity sensitivity. Statistical significance was defined as $p < 0.05$.

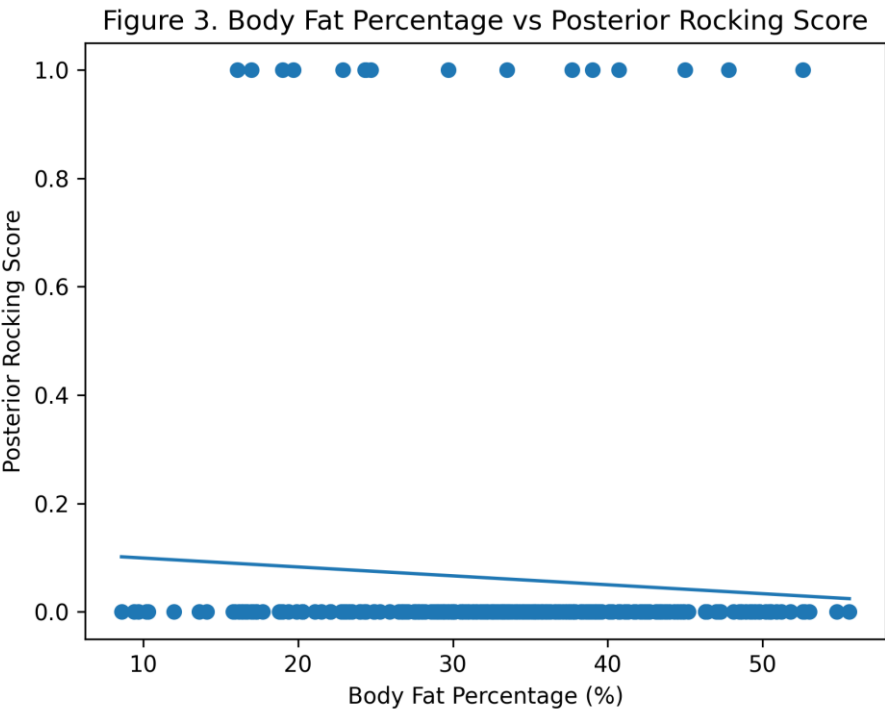
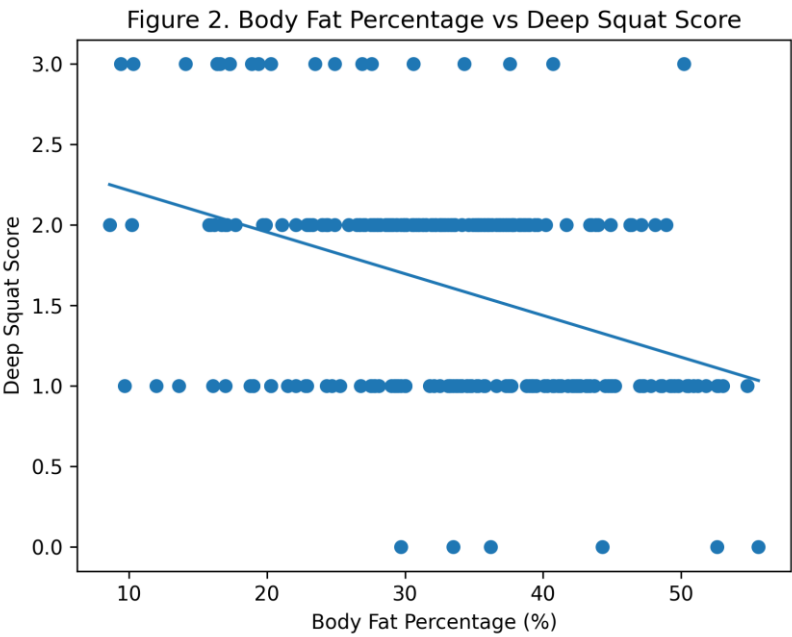
Descriptive Results

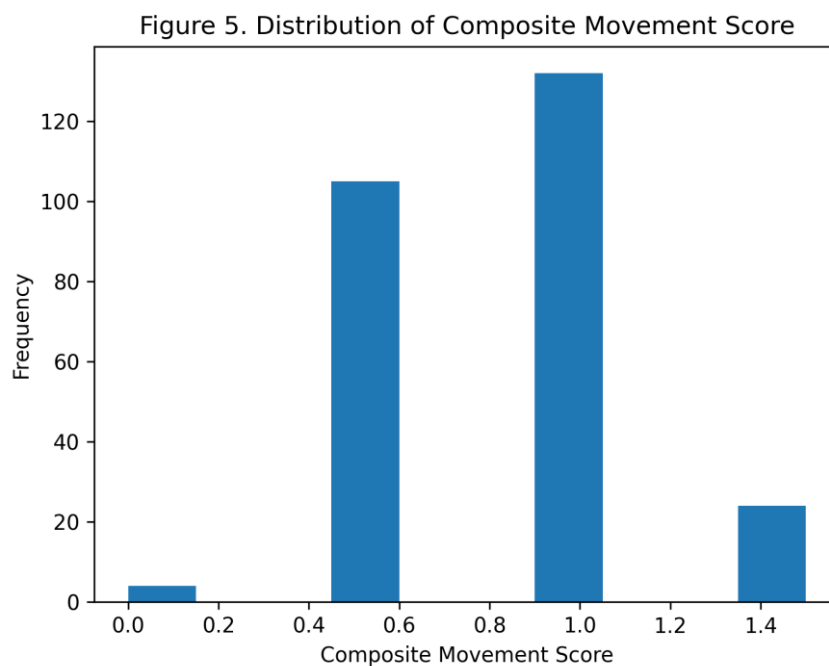
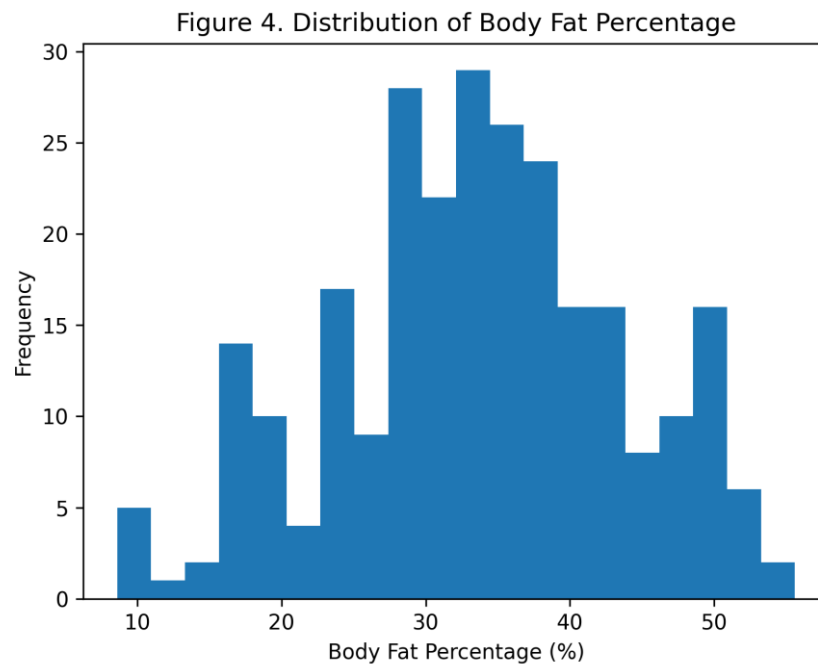
The sample included 265 adults (mean age 51.5 ± 12.2 years). Mean body fat percentage was $33.6 \pm 10.0\%$. Deep squat score averaged 1.60 ± 0.66 , while posterior rocking averaged 0.06 ± 0.24 . Composite movement score averaged 0.83 ± 0.33 .

Analytical Results

Body fat percentage demonstrated a moderate inverse association with composite movement score ($r = -0.417$, $p < 0.001$). Task-specific analysis revealed stronger association for the Deep Squat ($r = -0.391$). In multivariable regression, body fat percentage independently predicted movement score ($\beta = -0.0117$, 95% CI -0.0155 to -0.0080), with $R^2 = 0.208$.







DISCUSSION

The present findings demonstrate that adiposity exerts a statistically and biomechanically meaningful influence on functional movement competency. The stronger association observed for the deep squat aligns with mechanical load amplification theory¹⁰⁻¹⁴, wherein increased fat mass magnifies joint torque demands and reduces relative strength ratios.

Closed-chain multi-joint tasks likely amplify adiposity-related constraints through compounded compressive forces and altered center-of-mass dynamics.

Approximately one-fifth of movement variability was explained by adiposity, indicating clinically meaningful yet multifactorial determinants. Neuromuscular coordination, flexibility, training history, and motor learning likely contribute additional variance.

From a translational perspective, these findings support integration of DXA-derived body composition analysis within functional screening paradigms. Targeted adiposity reduction strategies may improve biomechanical efficiency and movement competency beyond cardiometabolic benefit. Future longitudinal studies should evaluate causal pathways and task-specific adaptation following body composition modification.

The present findings demonstrate that adiposity exerts a statistically significant and biomechanically meaningful influence on functional movement competency. Importantly, the magnitude of association observed between DXA-derived body fat percentage and composite movement score ($R^2 \approx 0.21$) suggests that approximately one-fifth of observable variability in movement quality may be attributable to adiposity alone. In complex biological systems such as human movement, where neuromuscular coordination, flexibility, strength, motor learning, and structural alignment all contribute simultaneously, explaining 20% of variance from a single physiological variable represents a clinically meaningful effect.

The stronger association observed for the deep squat compared with posterior rocking provides mechanistic insight into load-dependent adiposity sensitivity. Closed-chain multi-joint tasks require coordinated torque production across the hip, knee, ankle, and trunk segments under gravitational load⁶⁻⁹. Increased fat mass elevates compressive forces across weight-bearing joints and increases external moment arms, thereby amplifying muscular demand¹⁰⁻¹⁴. When relative strength (force normalized to body mass) declines, compensatory strategies may emerge, including reduced squat depth, increased trunk flexion, medial knee displacement, or impaired pelvic control⁶⁻⁹. These compensations likely explain the stronger correlation observed in the deep squat relative to posterior rocking, which places comparatively lower axial loading demands on the lower extremities.

Beyond mechanical loading, adiposity may influence neuromuscular performance through metabolic and inflammatory pathways. Adipose tissue functions as an active endocrine organ and is associated with low-grade systemic inflammation¹⁻⁵. Chronic inflammatory signaling has been linked to altered muscle activation patterns, reduced mitochondrial efficiency, and impaired fatigue resistance.

These physiological alterations may subtly degrade motor coordination during dynamic tasks, particularly those requiring sustained isometric stabilization or multi-planar control. Thus, the relationship observed in this study is likely multifactorial, reflecting both mechanical and metabolic influences.

It is also important to interpret these findings within the broader context of movement screening validity. Functional movement assessments are commonly used in athletic, military, and occupational settings to identify injury risk and movement dysfunction⁶⁻⁹. However, movement quality does not exist in isolation from morphological characteristics. The present findings suggest that adiposity may confound or partially mediate screening outcomes. Individuals with higher body fat percentage may demonstrate lower movement scores not solely due to motor deficits, but due to mechanical constraints imposed by increased mass and altered center-of-mass positioning¹⁰⁻¹². This distinction is critical when interpreting screening results in heterogeneous populations.

Interestingly, despite statistical significance, the majority of variance in movement competency remains unexplained by adiposity alone. This reinforces the multifactorial nature of movement quality. Strength capacity, mobility restrictions, proprioceptive acuity, training history, injury history, and motor learning all contribute meaningfully to functional performance. Therefore, while adiposity represents a significant determinant, it should be conceptualized as one component within a broader biopsychosocial framework of movement.

From a translational standpoint, these data suggest that body composition assessment should be considered when interpreting functional screening outcomes. Interventions targeting fat mass reduction may improve movement competency through improved relative strength ratios and reduced joint compressive forces¹⁰⁻¹⁴. Importantly, improvements in adiposity may yield dual benefit — cardiometabolic risk reduction¹⁻⁵ and enhanced biomechanical efficiency.

Future investigations should incorporate longitudinal and interventional designs to evaluate causality. Specifically, examining whether structured adiposity reduction programs produce measurable improvements in discrete movement patterns would provide valuable mechanistic clarity. Additionally, task-specific biomechanical modeling and regional fat distribution analyses (e.g., central vs peripheral adiposity) may further elucidate underlying pathways linking adiposity and movement quality.

STRENGTHS AND LIMITATIONS

Strengths

This investigation possesses several notable strengths. First, the use of dual-energy X-ray absorptiometry (DXA) provides a gold-standard measure of total body fat percentage^{15–17}. Unlike BMI, DXA differentiates lean and fat mass, reducing misclassification bias and allowing composition-specific inference. Second, the inclusion of discrete movement tasks in addition to a composite score allows mechanistic differentiation of adiposity sensitivity across movement patterns. This task-level analysis strengthens interpretability and enhances translational relevance.

Third, the analytical approach incorporated both bivariate correlations and multivariable regression adjusting for age and sex, allowing determination of independent predictive capacity. Adjusting for demographic factors is essential given known age-related neuromuscular decline and sex-based differences in fat distribution and strength characteristics. Finally, the sample size ($n = 265$) provides adequate statistical power to detect moderate associations and enhances stability of parameter estimates.

LIMITATIONS

Despite these strengths, several limitations warrant consideration. The cross-sectional design precludes causal inference. Although adiposity was independently associated with movement quality, temporal directionality cannot be established. It is plausible that reduced movement competency contributes to decreased physical activity and subsequent fat accumulation, suggesting bidirectional influence.

Second, movement scoring was based on ordinal assessment (1–3 scale), which, although standardized, may limit sensitivity relative to instrumented biomechanical analysis. Future studies incorporating motion capture or force plate analysis could provide greater resolution regarding kinetic and kinematic alterations.

Third, residual confounding variables such as training history, musculoskeletal injury history, flexibility measures, and strength testing were not included in the current model. Inclusion of these variables may further refine explanatory capacity. Additionally, regional fat distribution (e.g., visceral vs subcutaneous) was not analyzed and may differentially influence center-of-mass dynamics.

Finally, generalizability may be limited to community-dwelling adult populations. Replication in athletic cohorts, older adults, or clinical populations would enhance external validity.

CONCLUSION

In conclusion, DXA-derived body fat percentage independently predicts functional movement quality in adults, with approximately one-fifth of movement variability attributable to adiposity. Multi-joint, closed-chain tasks such as the deep squat demonstrate heightened biomechanical sensitivity to increased fat mass, likely due to amplified torque demands and altered center-of-mass positioning. These findings underscore the importance of composition-specific evaluation in functional movement screening and injury risk assessment.

Adiposity should not be viewed solely through a cardiometabolic lens; rather, it represents a meaningful biomechanical determinant of movement efficiency. Integration of body composition assessment into functional screening paradigms may improve interpretation accuracy and guide targeted intervention strategies. Future longitudinal and mechanistic investigations are warranted to determine whether reductions in adiposity translate into measurable improvements in movement competency and injury risk reduction.

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